

CO2 AT-3

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SUB:CSA1501 CLOUD COMPUTING

Q1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Answer

Auto-Scaling

Cloud auto-scaling automatically increases or decreases the number of servers based on demand. During seasonal sales, additional virtual machines are launched when CPU or network usage becomes high. After the sale, extra servers are removed to reduce costs.

Load Balancing

A load balancer distributes incoming user requests evenly across multiple servers. This prevents any single server from becoming overloaded and ensures smooth application performance.

Reduction of Downtime

- Prevents server overload.
- Automatically replaces unhealthy servers.
- Ensures continuous service during heavy traffic.
- Maintains high availability.

Improved User Experience

- Faster page loading.
- Reduced website crashes.
- Smooth checkout process.
- Better response time.

Role of Monitoring Tools

Monitoring tools continuously track:

- CPU utilization
- Memory usage

- Network traffic
- Response time
- Server health

If thresholds are exceeded, they trigger auto-scaling and alert administrators.

Real-World Example

During the **Amazon Great Indian Festival** or **Flipkart Big Billion Days**, millions of customers access the website simultaneously. Auto-scaling launches additional servers, while load balancers distribute traffic efficiently, preventing website failures and ensuring a smooth shopping experience.

Conclusion

Auto-scaling, load balancing, and continuous monitoring enable cloud platforms to efficiently handle unpredictable traffic spikes while reducing downtime and improving customer satisfaction.

Q2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Answer

Durability and Data Redundancy

Cloud storage providers store multiple copies of data across different disks, servers, and data centers. If one storage device fails, another copy is automatically available, ensuring high durability and reliability.

Encryption

- Encrypt data before storage.
- Encrypt data during transmission using SSL/TLS.
- Use AES-256 encryption for stored data.

Access Control

Use:

- Identity and Access Management (IAM)
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)

Only authorized users can access backup data.

Cost Advantages

Compared to traditional storage:

- No need to purchase physical hardware.
- Pay only for storage used.
- Lower maintenance costs.
- Automatic scaling without additional infrastructure.
- Reduced power and cooling expenses.

Disaster Recovery Example

A company's primary data center is damaged due to flooding. The latest encrypted backup stored in the cloud is restored to a new server, allowing business operations to resume quickly with minimal data loss.

Conclusion

Cloud storage provides secure, durable, and cost-effective backup solutions while ensuring fast disaster recovery through redundancy and encryption.

Q3. A mobile app company plans to deploy its application globally. Explain how Content Delivery Networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Answer

Content Delivery Network (CDN)

A CDN stores copies of application content on servers located around the world. Users receive data from the nearest server instead of the main data center.

Benefits

- Lower latency.
- Faster content delivery.
- Reduced bandwidth usage.
- Less load on the origin server.
- Improved application performance.

Edge Servers

Edge servers are geographically distributed servers positioned close to users.

Benefits:

- Faster response time.
- Reduced buffering.
- Improved gaming experience.
- Faster video streaming.

Global Availability

Cloud providers deploy services across multiple geographic regions and availability zones. If one region becomes unavailable, traffic is redirected to another healthy region.

Real-Time Example

Platforms like **Netflix** use CDNs to stream videos smoothly worldwide, while multiplayer games use edge servers to reduce lag and improve gameplay for users in different countries.

Conclusion

CDNs and edge servers significantly improve global application performance by reducing latency and ensuring reliable content delivery across multiple regions.

Q4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Answer

Regulatory Compliance

Cloud providers support industry regulations through:

- Compliance certifications.
- Security audits.
- Data protection policies.
- Access management.
- Audit reports.

Logging

Logging records:

- User login attempts.

- File access.
- Configuration changes.
- Financial transactions.

These logs support auditing and forensic investigations.

Monitoring

Monitoring continuously detects:

- Unauthorized access.
- Suspicious activities.
- System failures.
- Security threats.

Administrators receive real-time alerts for immediate action.

Encryption

- Encrypt stored financial data using AES-256.
- Protect data during transmission with TLS/SSL.
- Store encryption keys securely using a Key Management Service (KMS).

Data Sovereignty

To meet legal requirements:

- Store customer data only in approved geographic regions.
- Select cloud regions within the required country.
- Restrict cross-border data transfers.
- Regularly audit compliance.

Example

A bank processes online fund transfers. Transaction records are encrypted, access is controlled using IAM and MFA, all activities are logged for auditing, and customer data is stored only in data centers located within the country to satisfy regulatory requirements.

Conclusion

Logging, monitoring, encryption, and compliance controls help financial institutions protect sensitive information while meeting regulatory and data sovereignty requirements.

Q5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Answer

Containers vs Virtual Machines

Containers	Virtual Machines
Share the host operating system kernel	Each VM has its own operating system
Lightweight	Heavier due to full OS
Fast startup (seconds)	Slower startup (minutes)
Lower resource usage	Higher CPU and memory usage
Higher application density	Fewer VMs per server

Performance

Containers consume fewer resources, start quickly, and provide near-native performance, making them ideal for cloud-native applications.

Kubernetes Orchestration

Kubernetes simplifies deployment by:

- Automatically deploying containers.
- Scaling applications based on demand.
- Replacing failed containers.
- Load balancing traffic.
- Managing updates with minimal downtime.

Portability

Containers package the application and its dependencies together, allowing the same container image to run consistently on AWS, Azure, Google Cloud, or on-premises environments without modification.

Microservices Example

An **online shopping application** can be divided into independent microservices:

- User Service
- Product Catalog Service

- Shopping Cart Service
- Payment Service
- Order Service
- Notification Service

Each service runs in its own container and can be updated or scaled independently.

Conclusion

Containerization improves performance, reduces resource usage, and enhances portability. Kubernetes automates deployment and scaling, making containerized applications highly efficient across multiple cloud platforms.